



Sunlight-Driven Photocatalytic Degradation of Methylene Blue with Facile One-Step Synthesized Cu-Cu₂O-Cu₃N Nanoparticle Mixtures

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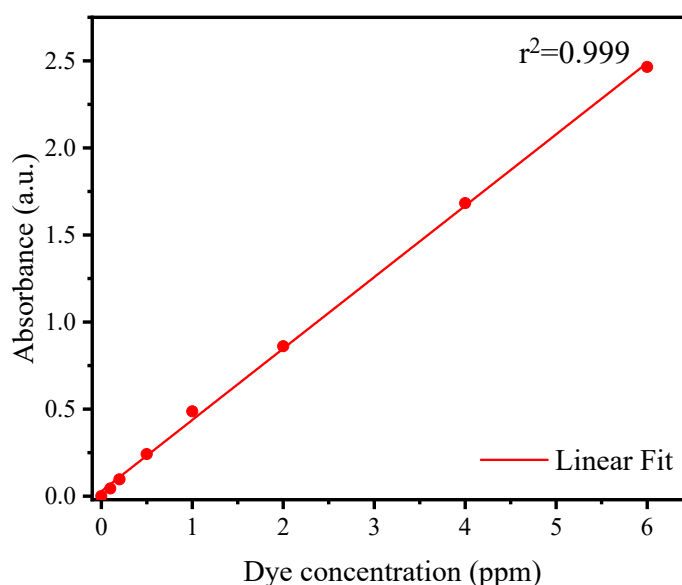
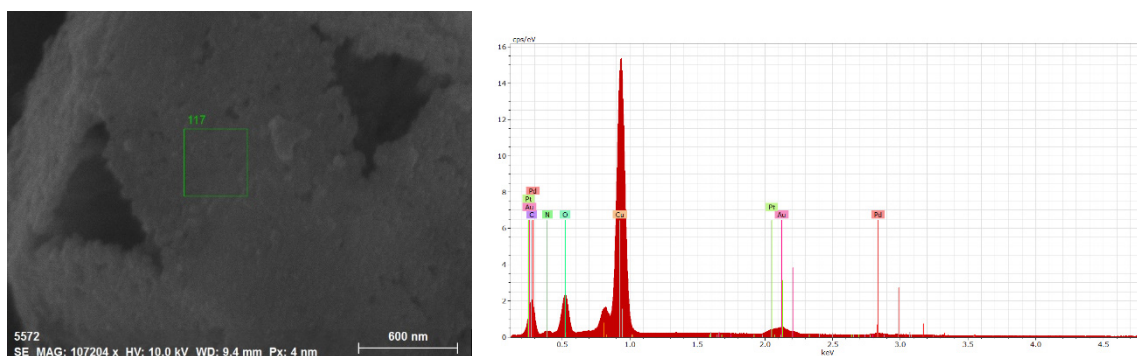


Figure S1. Calibration curve of MB using Lambert-Beer's law obtained with Lovibond photometer at a wavelength of 660nm.

For the calibration curve, eight samples at different concentrations of 6, 4, 2, 1, 0.5, 0.2, 0.1 and 0 mg/L of MB were prepared. The absorbance of the solutions was measured using a Lovibond photometer with a wavelength of 660nm and plotted as a function of MB concentration.



Element	Series	unn. [wt.%]	C norm. [wt.%]	Atom. [at.%]	Error (1 Sigma) [wt.%]
Copper	L-series	89.88	89.88	68.66	9.52
Oxygen	K-series	8.66	8.66	26.26	1.04
Nitrogen	K-series	1.46	1.46	5.08	0.26
Total:		100.00	100.00	100.00	

Figure S2: EDX of the truncated octahedron in sample 6h.

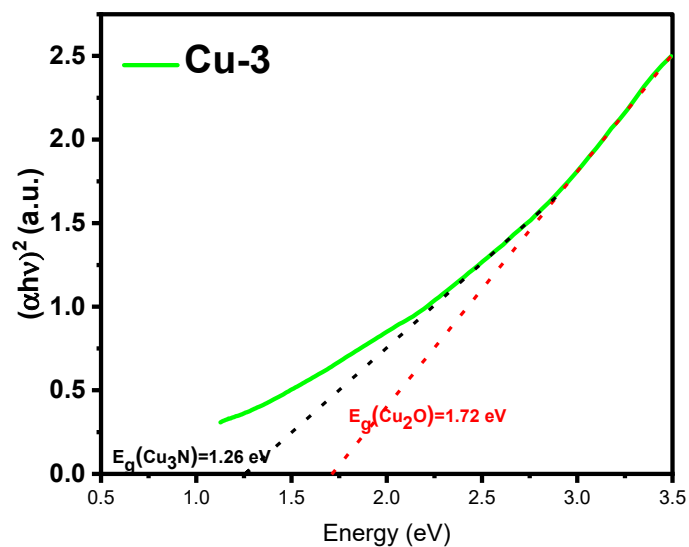


Figure S3. Tauc plots of Cu-3 obtained from UV-Vis absorption spectrum of the sample. The extrapolation of the linear portions of the curves provide the band gap of the Cu_3N and Cu_2O phases.

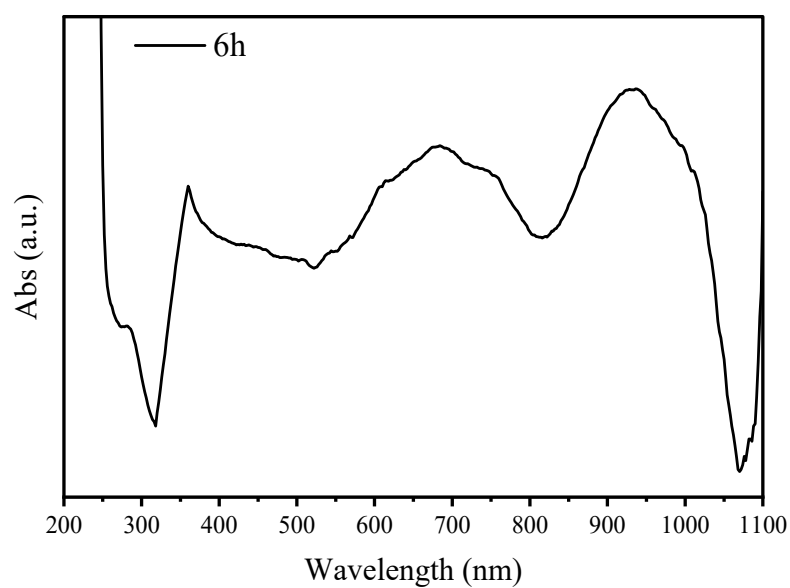


Figure S4. UV-Vis absorption spectrum of 6 h sample showing the absorption peak at ~300 nm.

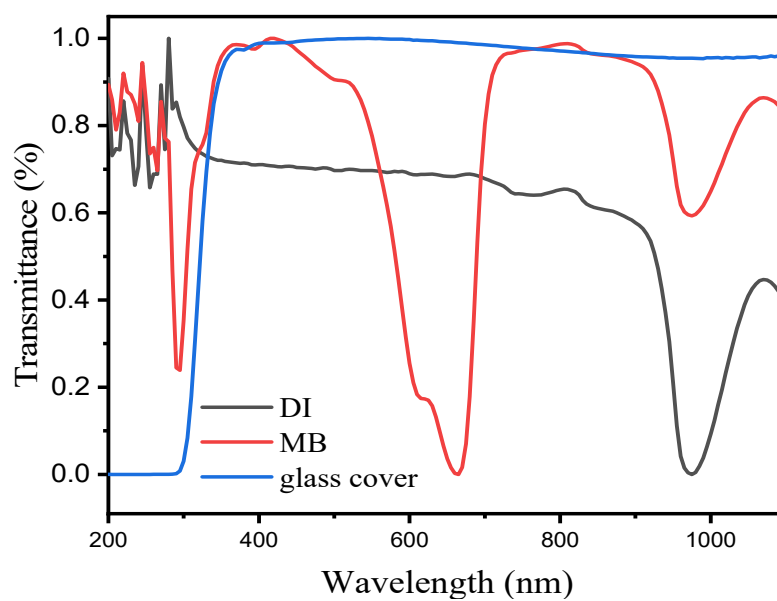


Figure S5. S. UV-Vis spectra of distilled water (DI), Methylene Blue (MB) and glass cover.